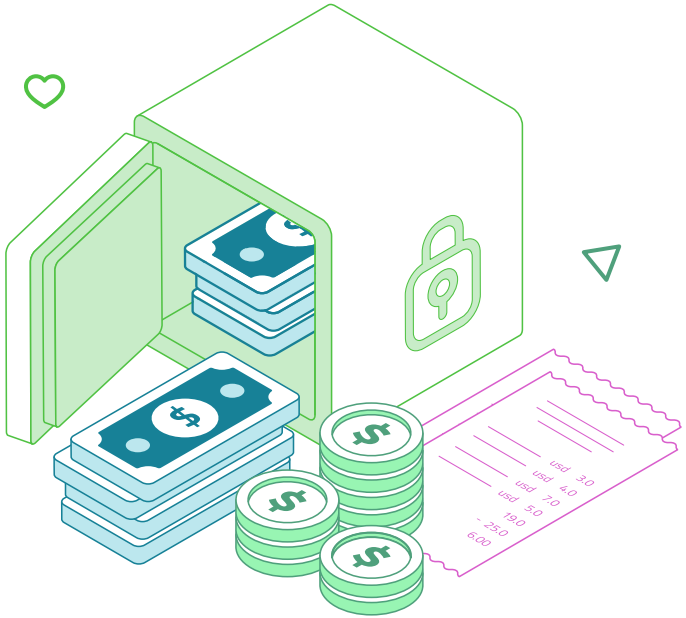

Small Business Loan Risk

Project Lead: Philbert Fan
Members: Jessica Kim, Kevin Tran, Sarah Dias, Chloe Lee, Lucy Guo

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Project Problem and Introduction

01

Predicting Credit Risk & Loan Defaults

Credit Risk: The chance that someone will not pay back the money they borrowed

Why it matters?
Bank lends money to make money
If borrowers not pay back lose money

End Goals

Based on **past data,** will this borrower probably **pay back the loan?**

- Create a model through **supervised learning** & **classification machine learning** techniques

EDA and Pre-processing

02

Exploratory Data Analysis (EDA)

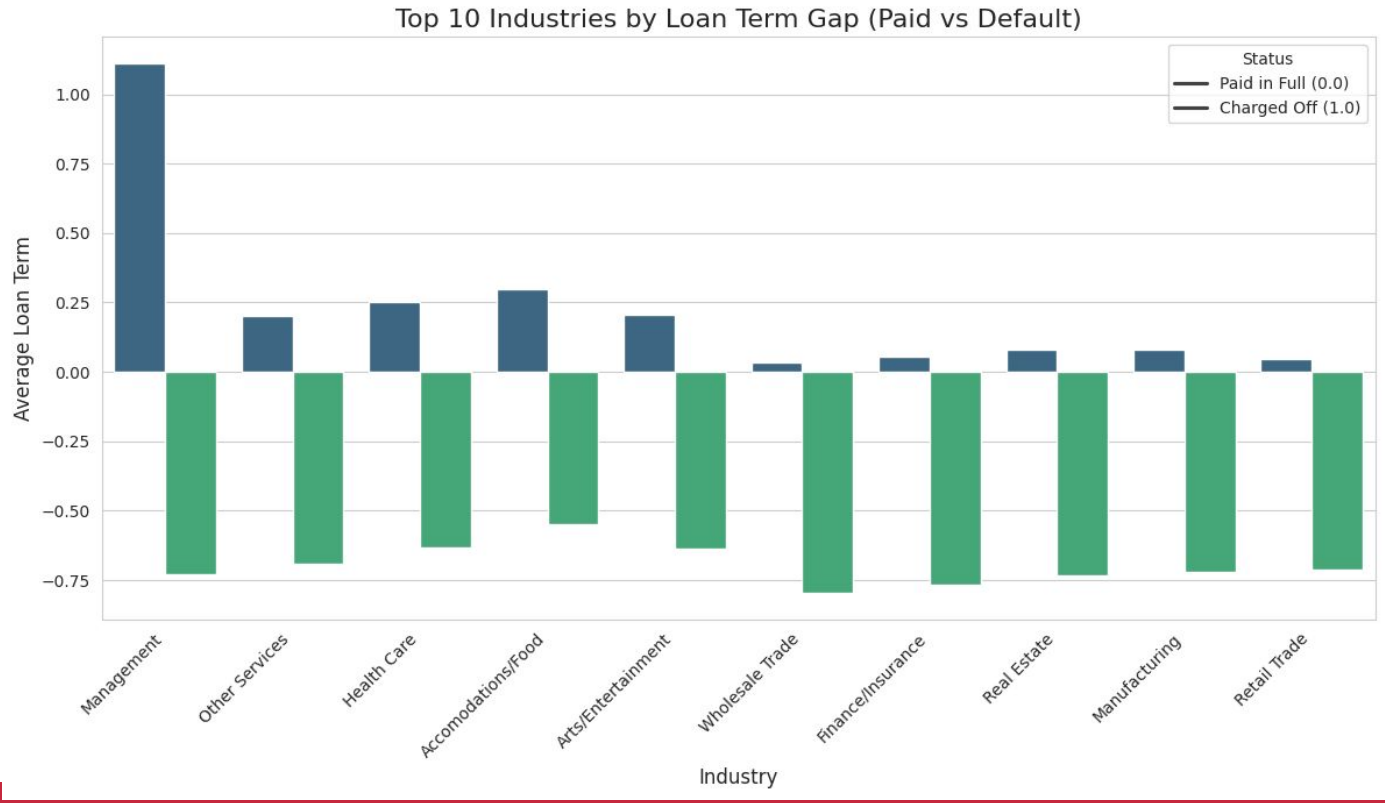
Dataset: Comprehensive record of U.S. Small Business Administration (SBA) loan data spanning from 1987 to 2014. It tracks the lifecycle of loans guaranteed by the federal government to support small business growth.

- 899,164 observations and 27 variables
    - Variable Categories: Business Identifiers, Operational Details, Loan Specification, Government Guarantee, and Outcome Label

Data Analysis:

- Loan Term most correlated with loans paid in full and Fiscal Year of Commitment most correlated with loan defaulting
- Most of our numeric variables like, number of employees, jobs created, and jobs retained were extremely skewed due to high value outliers

EDA Visualization



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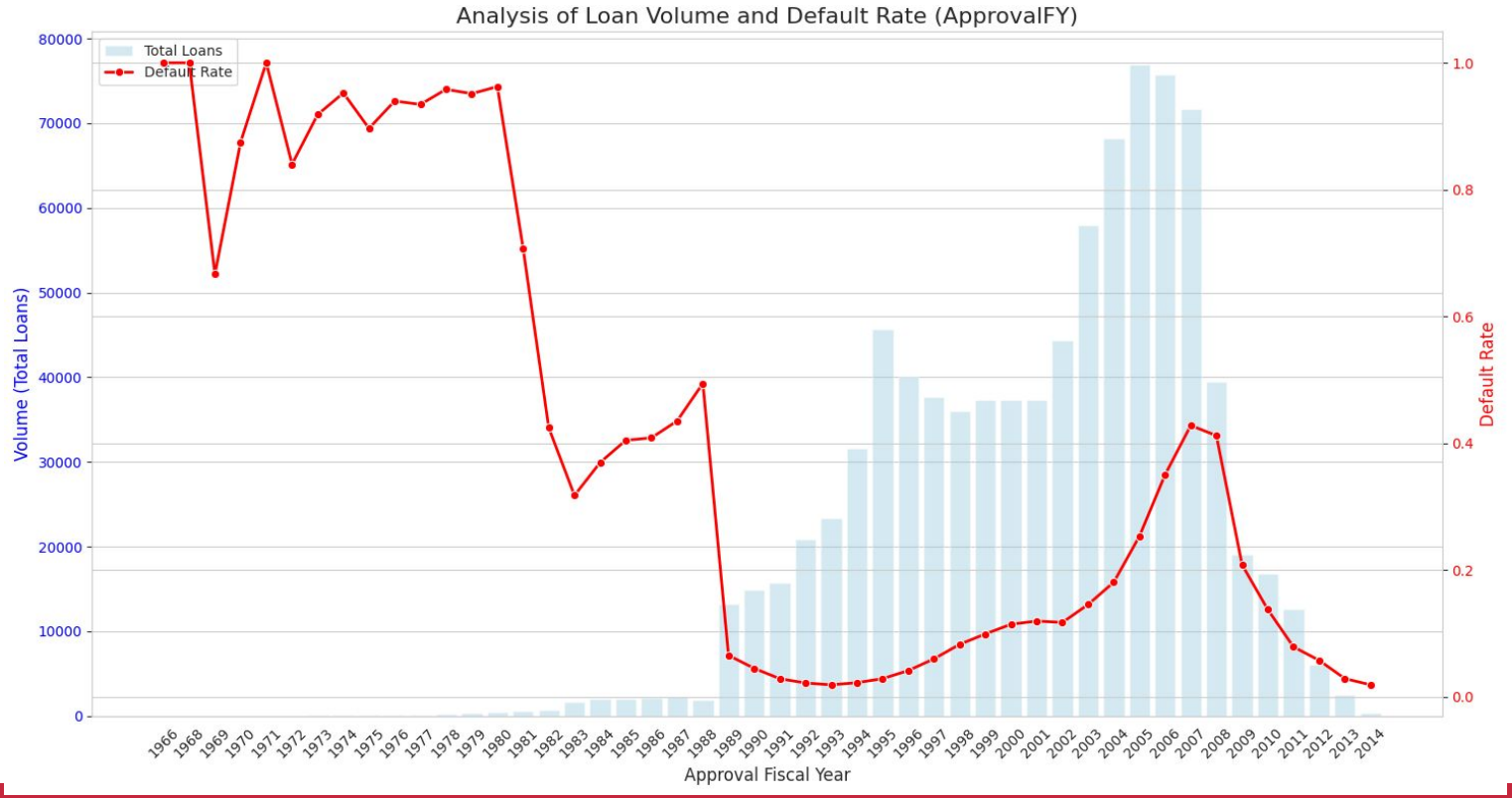

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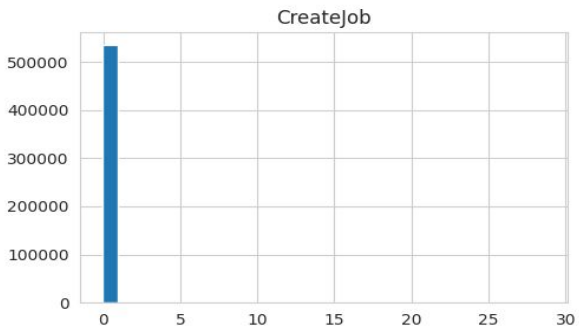
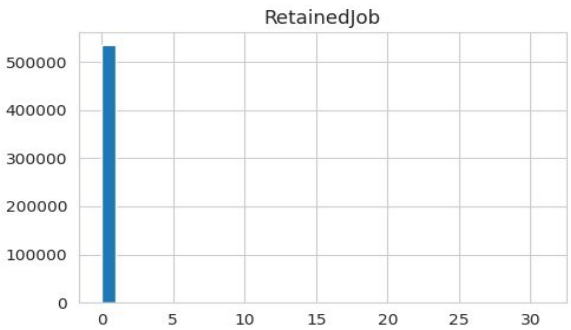
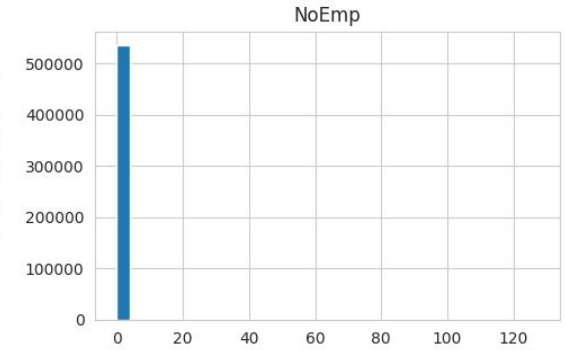
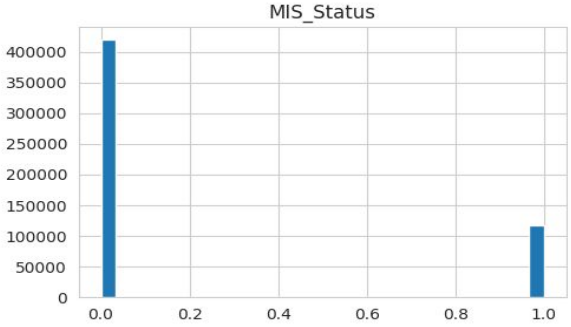
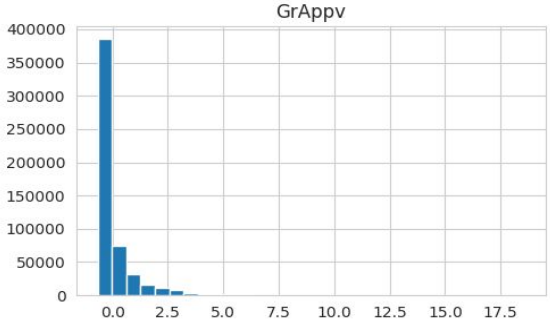










EDA and Preprocessing

- Class Imbalance in Target Variable: ~82% Paid in Full and ~17.5% Charged Off
- High Noise Level: Lots of unique irrelevant/redundant variables and data leakage (i.e. Business name, city, state, Loan#, ApprovalDate, ChgOffDate, Gross Disbursement)
- Missing Values: ~82% of ChgOffDate variable missing
- Data Types:
    - Numeric variables classified as string type (GrAppv & SBA_Appv)
    - Identifier/Nominal variables classifies as numeric type (LoanNr_ChkDgt, NAICS code, & zip code)
    - Binary variables contained more than 2 values and classified as type string (MIS_status, lowDoc, UrbanRural)

How We Handled Our Dataset

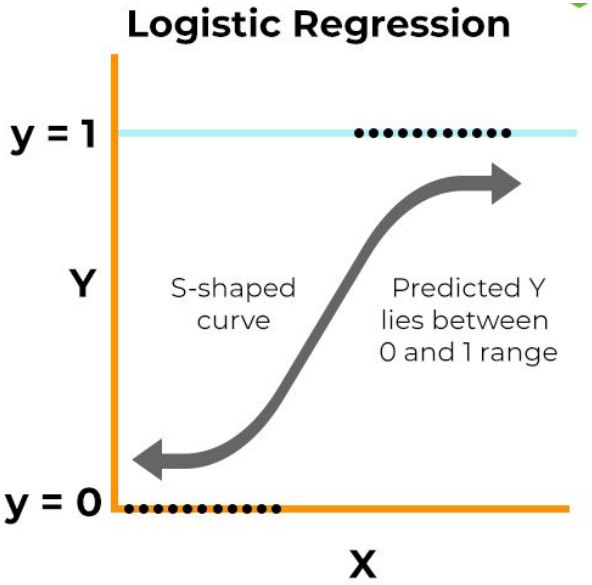
- Class Imbalance in Target Variable: ~82% Paid in Full and ~17.5% Charged Off
  - Used Class Weight = ‘balanced’ in our model later
- Dropped redundant columns including ChgOffDate, ApprovalDate, LoanNr_ChkDgt, City, State, Gross Disembursement, and Name
- Converted GrAppv & SBA_Appv to numerical datatypes
- Converted the NAICS code to categorical datatype
- Numerical values were scaled
- Used one-hot encoding for our categorical variables
  - Converted MIS_status, lowDoc, UrbanRural into binary values

ML Models

03

Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for binary classification by modeling the probability of an event using a sigmoid function



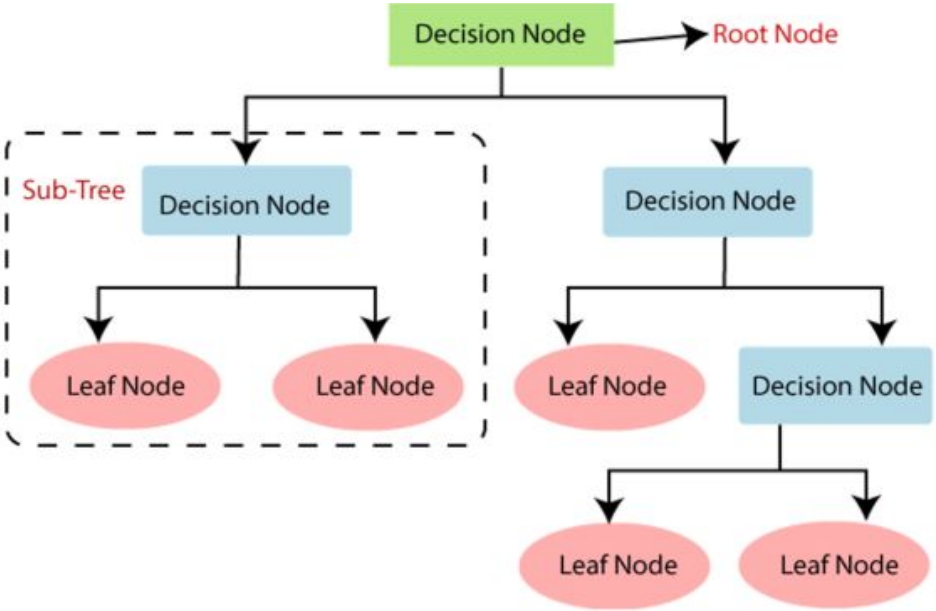
https://www.researchgate.net/figure/Decision-tree-illustration-Source_fig2_367234728

$$P(Y=1) = \frac{1}{1+e^{-(\beta_0+\beta_1 x_1+\ldots+\beta_k x_k)}}$$

- Model guesses initial coefficients.
- It calculates predictions using the sigmoid function
- It compares predictions to actual values(0 or 1).
- It calculates error using log loss algorithm
- It adjusts coefficients to reduce error.
- Repeats this process many times.

Decision Tree Classification

Decision Tree Classification is a non-parametric supervised learning algorithm that maps a series of nested "if-then" logical splits into a flowchart structure to predict the category of a data point.

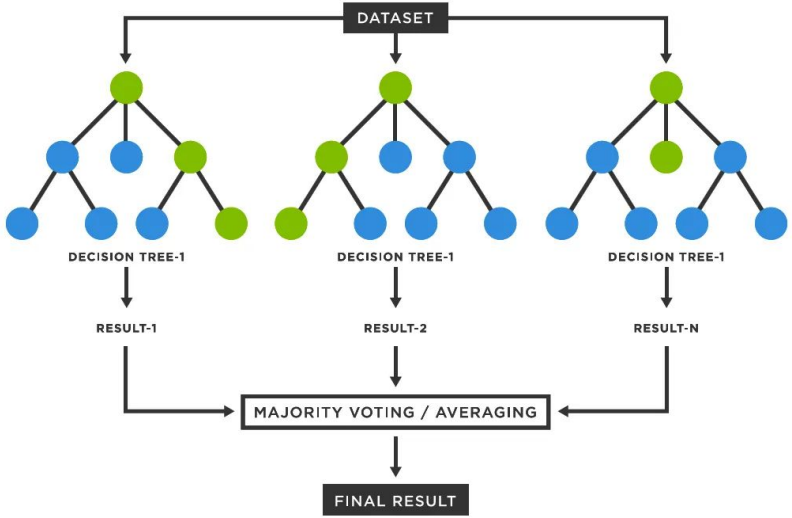


https://www.researchgate.net/figure/Decision-tree-illustration-Source_fig2_367234728

- Split data based on Gini impurity or entropy to predict most predictive features/variables
- Utilize nonparametric logic to capture complex, nonlinear relationships between variables
- Tune tree depth and node/leaf size to prevent overfitting and underfitting.

Random Forest Classification

Random Forest is an ensemble learning method that constructs multiple decision trees using bootstrap aggregation and random feature selection, and outputs the majority class prediction in classification tasks.

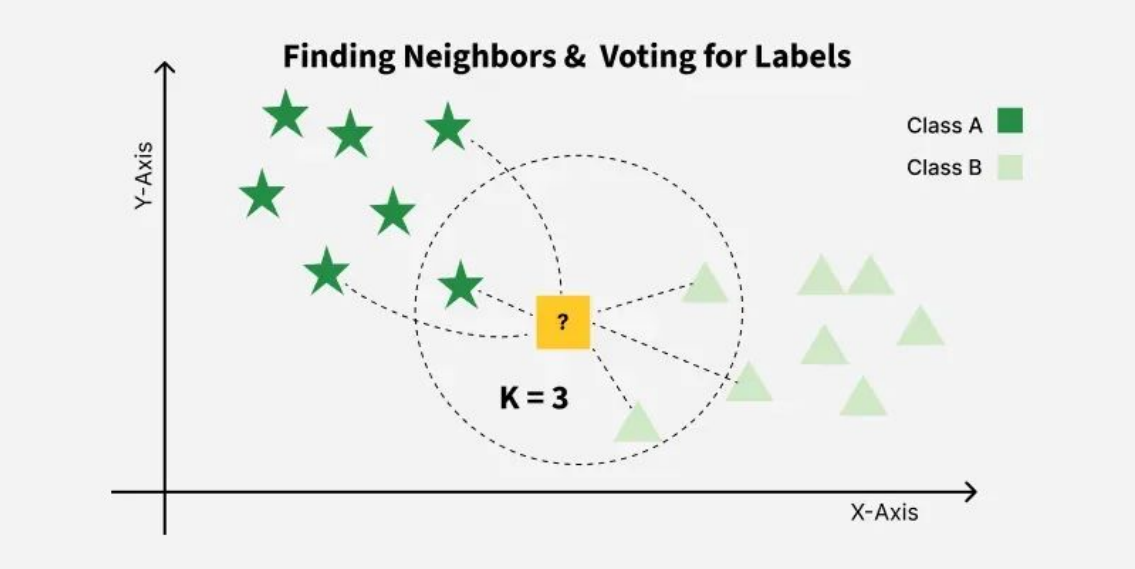


https://blog.devops.dev/a-deep-dive-into-the-random-forest-algorithm-d4a8950507a3

- Preprocess data: encode categorical variables, split into train/test sets, and handle class imbalance using class_weight='balanced'
- Build a baseline Random Forest classifier to establish initial performance (F1-score benchmark)
  Tune hyperparameters (n_estimators, max_depth, min_samples_split, min_samples_leaf) using RandomizedSearchCV with cross-validation
- Select the best-performing model based on cross-validated F1-score and evaluate on the test set

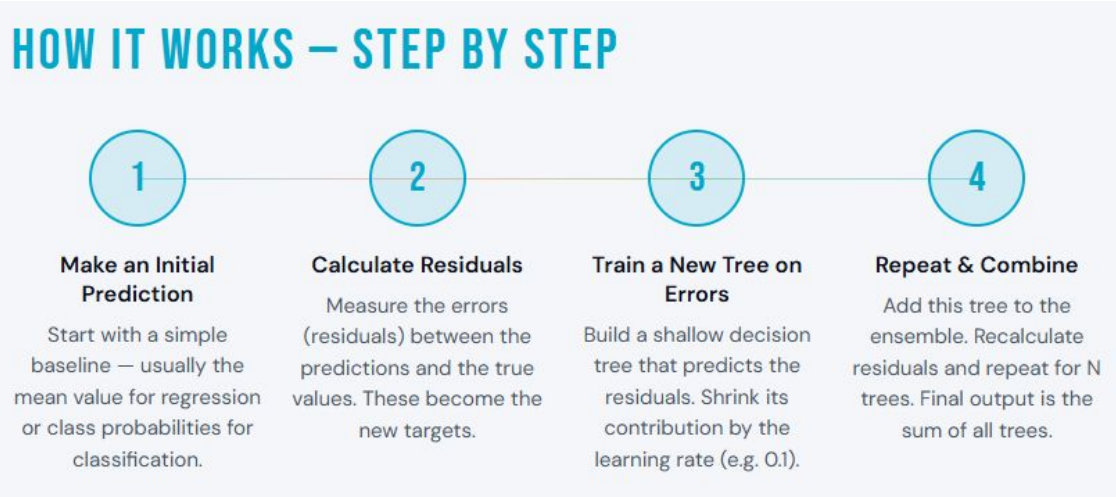
K-Nearest Neighbors (KNN) Classification

Distance-based algorithm that classifies a data point based on their proximity to the K closest training examples, assigning it the majority class



- Scale and select relevant features
- Choose appropriate distance metric
- Tune k (number of neighbors) using cross-validation to maximize F1/recall

XGboost Classification



XGboost builds many decision trees in sequence, where each new tree is trained to correct the errors of the previous trees combining the predictions into one powerful model.

In order to prevent overfitting, XGboost uses lasso and ridge regularization, which adds a penalty that discourages the model from relying too heavily on any one feature.

Model Comparisons

04

Evaluation Metrics

True Positive: Predicted loan to be defaulted and it defaulted | **True Negative:** Predicted borrower would pay and they paid
False Positive: Predicted to default, but actually paid | **False Negative:** Predicted a borrower to be good, but they defaulted

Precision: Ratio of True Positives to total number of positive predictions → TP / (TP + FP)

- Minimize False Positives

Recall: Ratio of True Positives to total number of actual positives → TP / (TP + FN)

- Minimize False Negatives

F1 Score (most important): harmonic mean of Precision and Recall

- Used for uneven class distributions
- Finds balance between avoiding FP and catching all actual positives
- Punishes extreme Precision or Recall values

Comparing Models Using F1 Scores

MODEL COMPARISON — F1-SCORE

Best model · > 0.80 · > 0.70 · < 0.70

MODEL	MEAN F1-SCORE	STD DEV
Random Forest	0.8374	0.0081
Logistic Regression	0.6608	0.0085
XGBoost	0.8733	0.0109
Decision Tree	0.8081	0.0085
KNN	0.7041	0.0081

Best Model by F1-Score: **XGBoost**

The best model: XGBoost

- Works great with tabular/structured data
- Performs well with complex non-linear feature interactions
- Precision and recall higher than linear and distance-based models
- Most efficient at identifying positive cases while minimizing false positives and false negatives

Final Model Insights

05

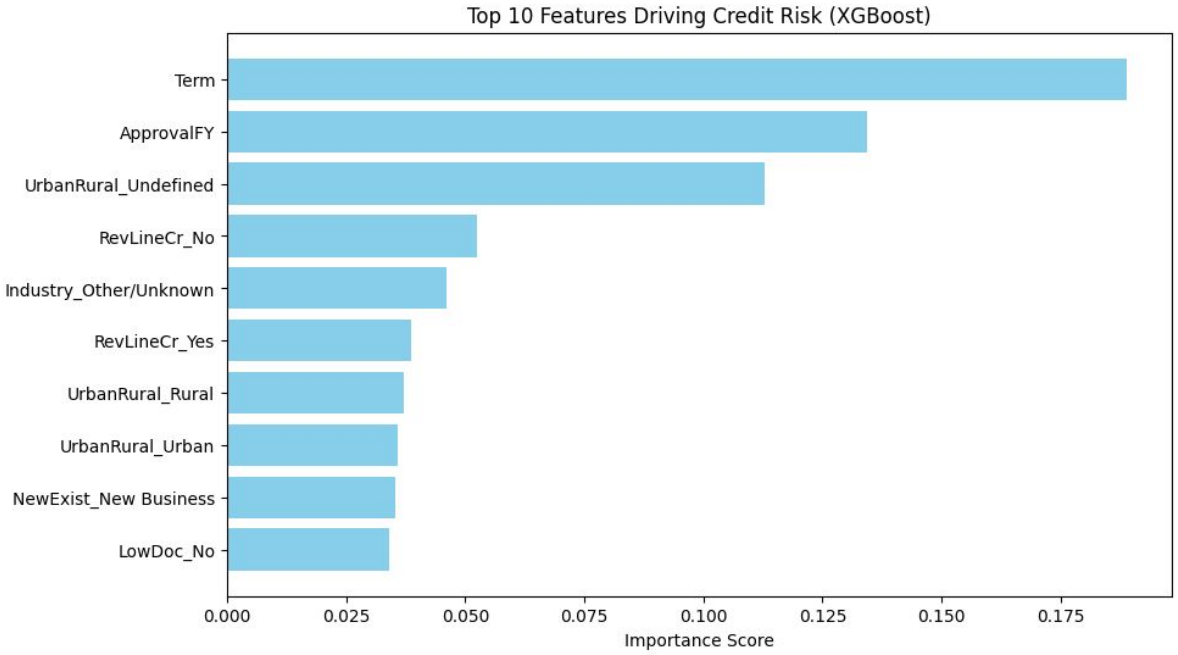
Model Tuning

After determining that **XGBoost** would be the best performing model, we decided to find the best parameters for the model.

We created a parameter grid of possible values and then used Randomized Search CV to score 10 random combinations by F1 score. After identifying the highest-scoring combination, we fit the given parameters into XGBoost to create our final model.

```
final_model = XGBClassifier(eval_metric='logloss',
                            subsample = 0.8,
                            n_estimators = 100,
                            max_depth = 6,
                            learning_rate = 0.1,
                            gamma = 5,
                            colsample_bytree = 0.8,
                            random_state = 42)
```

Feature Importance



Feature importance reveals how important a variable is in predicting the given outcome (in this case loan default).

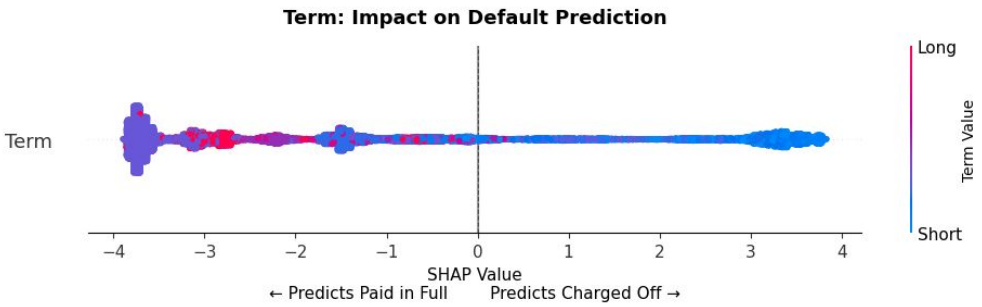
Among the features, the most important are **Term (Loan term in months), ApprovalFY (Fiscal year of approval)**. Surprisingly, having an undefined value for urban_rural seems to have predictive value for this dataset although it may not be useful in real-world context.

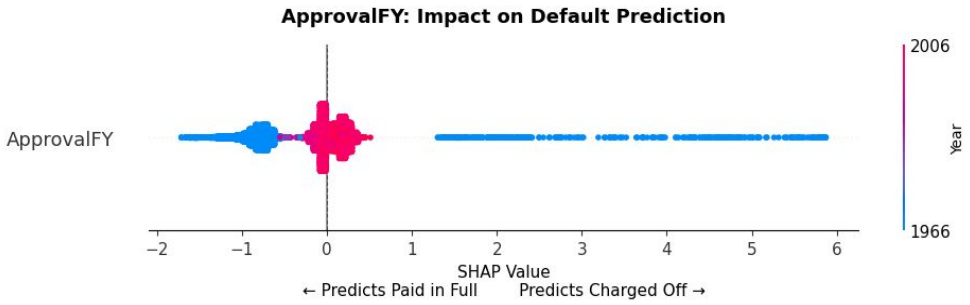
SHAP - A deeper look at Feature Impact

SHAP (SHapley Additive exPlanations) can show how each variable contributes to a prediction by pushing it toward either default or paid in full. Let's revisit the top two variables to see how they impact prediction!

Term - Longer loan terms increase the likelihood of default, while shorter terms tend to push predictions toward loans being paid in full.

ApprovalFY - Earlier approval years (older loans) predict default. More recent approvals (red) seem safer, possibly reflecting tighter post-crisis lending standards.

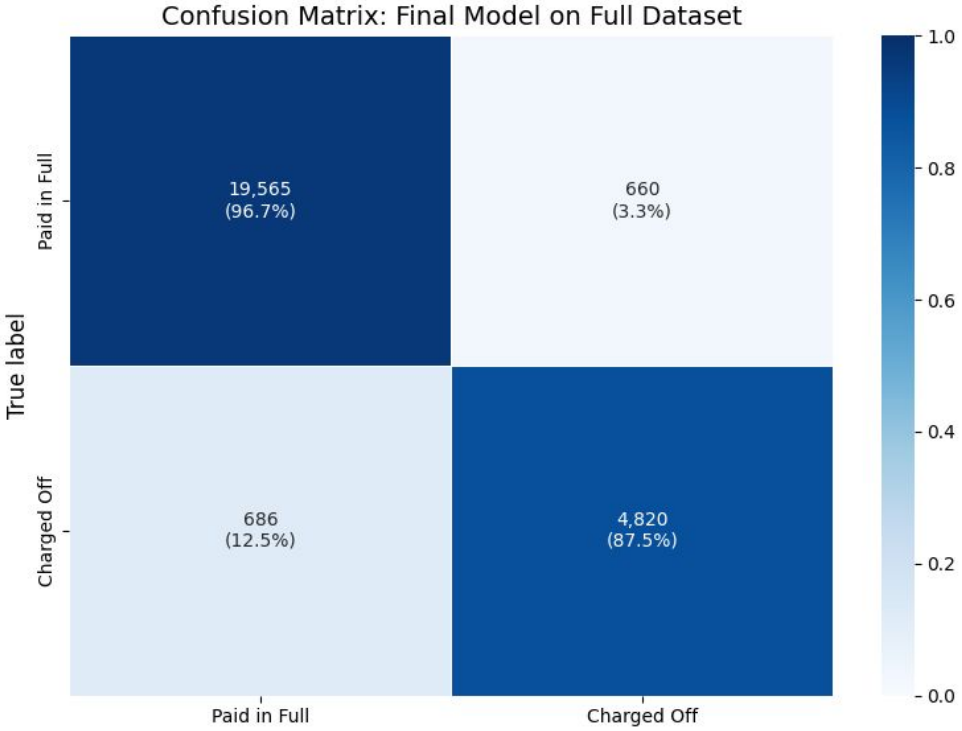




Confusion Matrix

This confusion matrix shows the breakdown of correctly and incorrectly guessed loan outcomes. We can see that if a loan will truly be paid in full, the model has a **96% chance** of correctly guessing that it will be paid in full. However, if a loan will truly be charged off, the model has an **87% chance** of correctly guessing that it will be charged off.

This difference in accuracy may be due to the fact that this is an **unbalanced dataset** => more paid in full loans means more training signal, giving the model stronger predictive power for this class.



Summary

06

Summary of Our Process

- Selected the **SBA Loan Dataset** as our primary dataset for credit risk prediction
- Conducted structured EDA to understand data quality, imbalance, and risk patterns
- Cleaned and preprocessed the data (handled missing values, fixed data types, etc.)
- Engineered meaningful features and validated assumptions
- Trained and tuned multiple classification models using cross-validation
- Evaluated models using Precision, Recall, and F1-score

Our insights from EDA

- The dataset is highly imbalanced
- Some variables required careful auditing to avoid target leakage (post-outcome information)
- Many missing values in our dataset (82% in ChgOffDate variable, etc.)
- Inconsistent datatypes

Summary of Insights from Model Interpretation

- By comparing the F1-scores and some other evaluation metrics across all models, **XGBoost** demonstrated the best overall performance.

- Default risk is strongly influenced by loan term and fiscal year of approval

- Shorter loan terms have a higher impact of whether a loan will be paid in full whereas longer loan terms suggest chances of default

- Earlier approval years were associated with more default, while recent approvals seem safer, suggesting possible tighter lending standards

Thanks for Listening!